

Neil Maushard

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EDUCATION

University of Illinois at Urbana-Champaign (UIUC), Grainger College of Engineering

Expected Dec 2026

Bachelor of Science in Mechanical Engineering, Minor in Electrical Engineering

GPA: 4.0/4.0

Related Coursework:

Robot Dynamics and Controls | Mechatronics | Mechanical Design | Advanced Dynamics | Signal Processing | Solid Mechanics

TECHNICAL SKILLS

Mechanical/Electrical: SolidWorks, Fusion 360, 3D Printing, Laser Cutting, Altium, KiCAD, PCB Design, CNC

Programming: Python, C, MATLAB/Simulink, Embedded Programming, OpenCV, rospy, C++, HTML, Bash, Git, Linux

Robotics: ROS, SLAM, Motor Control Schemes, Path Planning, Raspberry Pi, Kalman Filtering, Foxglove, LabVIEW

EXPERIENCE

Vermeer, Robotics Engineering Intern

May 2026 – Aug 2026

- Designed a scalable system to validate multiple visual navigation devices per drone flight, reducing validation time by 75%
- Programmed ROS software stack in Python to aggregate data from many devices and publish it to a central Foxglove visualization
- Streamlined startup via custom executable to manage environments and multiprocessing while dynamically resolving ROS IPs
- Configured a custom network using a router for DHCP and switch for scalability to broadcast data remotely with a wireless radio
- Designed and fabricated laser-cut carbon fiber mounting hardware and 3D printed electronics housing in Fusion 360

MechSE @ UIUC, Course Assistant - ME 370 (Mechanical Design)

Jan 2026 – May 2026

- Developed interactive homeworks using HTML and Python, implementing random parameter generation and robust grading logic
- Authored problem sets covering design of linkages and mechanisms, dynamics, kinematics, motors, gears, and more

KIMLAB, Research Assistant

Jan 2026 – May 2026

- Contributed to the development and design of Ringbot, a monocyte robot with leg modules that drive, turn, balance, and stand
- Prototyped and iterated printed components for Ringbot using SolidWorks, resolving interferences and high-friction joints
- Integrated gearing, actuators, and sensors into leg modules, and fabricated wire harnessing for reliable power and data routing

Hendrickson, Manufacturing Engineering Intern

June 2025 – Aug 2025

- Eliminated need for inefficient part-marking tool by integrating part-marking into pre-existing laser cutting programs
- Characterized springback of a novel material relative to aluminum and steel, improving tolerances and reducing scrap
- Designed tooling (vacuum, magnetic, chain-driven) to improve ergonomics and efficiency in handling heavy parts

Hutchens Research Group, Research Assistant

Jan 2024 – May 2025

- Designed an experimental setup using SolidWorks and FEA simulations to improve data collection and ensure stability
- Fabricated custom components via 3D printing and laser cutting, performing iterative stress testing to resolve failure modes
- Analyzed and visualized data using Python and MATLAB to evaluate performance and drive design decisions with supervisors

PROJECT HIGHLIGHTS (see more at [neiljm2.github.io](https://github.com/neiljm2))

SLAM Golf Putting Robot

April 2026 – May 2026

- Programmed a robot car using a finite state machine in C to map an arena with obstacles, a ball, and a goal to shoot the ball at
- Implemented SLAM by fusing LiDAR, IMU, and encoders to generate a real-time 2D occupancy grid to track the robot's location
- Designed a potential field path planner to navigate the arena and used blob detection and AprilTag vision to locate the ball/goal
- Built a real-time map in LabVIEW for real-time visualization of the arena with the obstacles, ball, goal, and robot's pose estimation

Robotic Arm Obstacle Course

April 2026 – May 2026

- Developed a control scheme in C for a 4-DOF arm to navigate an obstacle course using task-space PD control & impedance control
- Designed a waypoint-based linear interpolation trajectory generator for smooth, constant-velocity 3D motion without overshoots
- Implemented adaptive task-space impedance control with frame transformations to tune directional compliance for precise tasks

6-Legged Walking-Dispensing Robot

Sep 2025 – Dec 2025

- Designed a legged robot with a custom powertrain, using a single DC motor to walk 6 m/min and dispense a payload every 2m
- Performed simulations of custom 4-bar linkage leg designs to refine the robot's stride and achieve the desired walking gait
- Analyzed walking-motion kinematics and dynamics to smooth acceleration profiles and lower transient force spikes

6-Axis Robot Arm, Electrical Team Lead

Sep 2024 – April 2025

- Led 10 engineers to integrate electronic systems into a 6-DOF robotic arm with a \$300 budget for UIUC's Engineering Open House
- Built an Arduino-based stepper motor control system that converts inverse kinematics results into 0.1°-accurate positioning
- Oversaw development of a closed-loop PID DC motor control system using motor controller, encoder, and Arduino
- Designed the full electrical architecture and collaborated with mechanical team to integrate 25 components and route 60ft of wire